

Structure Aware Chunking for Scientific Retrieval

Abstract

We study how document structure affects retrieval quality over machine learning papers. We find that respecting section boundaries matters more than chunk size across every configuration we evaluated.

1 Introduction

LEFTONE Retrieval augmented generation systems depend on how documents are divided into passages.

LEFTTWO A poor chunker cannot be repaired by a better retriever.

LEFTTHREE Section boundaries carry meaning that fixed windows discard.

2 Related Work

RIGHTONE Prior work has examined fixed size chunking extensively.

RIGHTTWO Few studies measure the effect of section boundaries directly.

RIGHTTHREE We address that gap with a measured ablation over both factors.

3 Method

We parse each document with a layout aware extractor and detect headings from font size and numbering.

3.2 Training

We train for ten epochs with a batch size of 512 sequences on eight accelerators.

4 Results

Recall at five improves from 0.61 to 0.78 when section boundaries are respected.

Chunk size alone accounts for less of the difference than we expected.

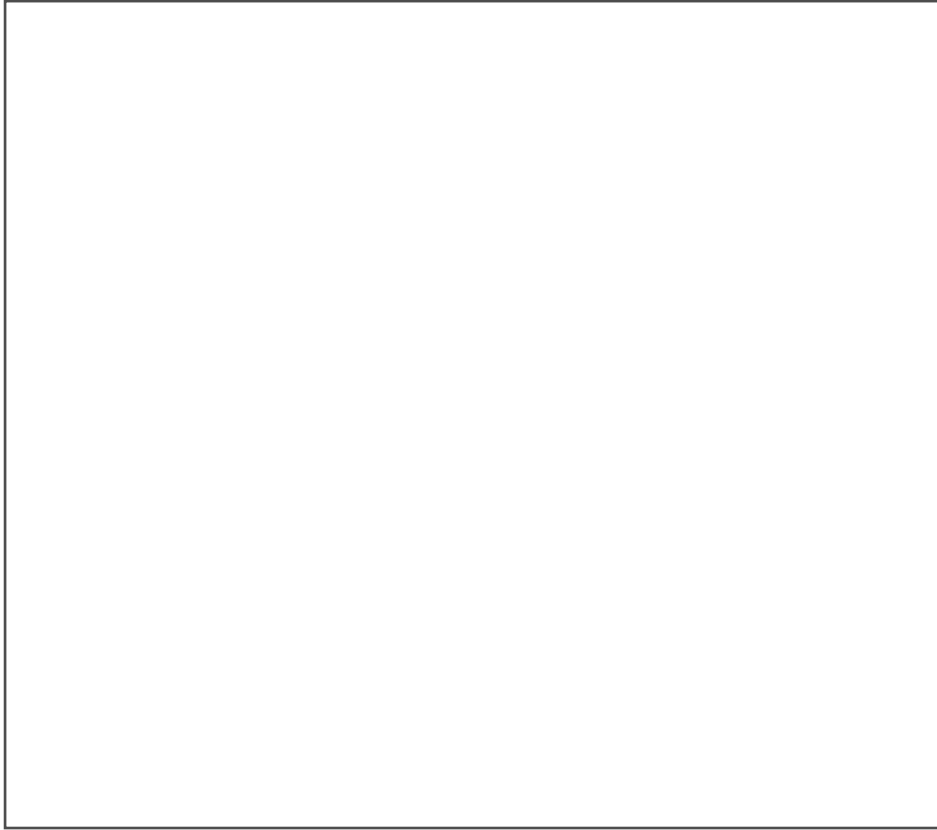


Figure 1: Recall against chunk size.

References

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